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Prof. David E. Manolopoulos
Editor-in-Chief
The Journal of Chemical Physics
American Institute of Physics

Dear Prof. Manolopoulos,

We submit our manuscript, “Suppression of Pure Dephasing and Enhancement of Excitonic Coherence via Non-Markovian Spectral Bath Engineering,” for consideration as an Article in The Journal of Chemical Physics.

What we did: Understanding how complex condensed-phase systems dissipate energy requires rigorously mapping non-Markovian system-bath correlations. In this work, we deployed the exact Process Tensor Hierarchy of Pure States (PT-HOPS) algorithm to investigate how arbitrary manipulations of the *external* incident photon bath modulate internal electronic dynamics in the Fenna–Matthews–Olson (FMO) complex. By subjecting the matrix to engineered dual-band non-Markovian environments (750 nm and 820 nm), we successfully uncoupled the excitonic network from the destructive thermal background noise.

Theoretical insights: Structurally, we prove that customized spectral transmission filters behave as active polaron-dressing operators. The engineered bath mathematically truncates the effective reorganization energy without altering the resonant Franck-Condon inter-site overlaps. As a direct consequence, the pure dephasing framework (Γ_{pd}) is forcefully suppressed.

Physical Outcomes: Utilizing a full 295 K Drude-Lorentz foundation interlaced with explicit vibronic underdamped sub-oscillators, these polaron mechanisms produced a 50 % temporal extension of quantum superpositions *in vivo*, generating an aggregated 25 % leap in target trap transfer populations completely robust against 50 cm^{-1} static deformation matrices.

Why JCP: This manuscript explicitly addresses the core mission of JCP: exposing the fundamental physical frameworks governing complex chemical behaviors. By merging exact theoretical benchmarking against HEOM algorithms with profound non-equilibrium transport insights via polaron me-

chanics, we provide the foundational chemical physics blueprint for deploying driving fields as proactive quantum control levers.

All authors explicitly approved this manuscript. No part of this work is under review elsewhere. We declare no competing financial interests.

Thank you for your consideration.

Sincerely,

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